Karthick Balakrishnan

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LinkedIn | Portfolio

Professional Summary

- 15+ years of UX/product design experience across enterprise systems, fintech, manufacturing, energy & public sector. Expertise in AI-powered UX, LLM-enhanced workflows, and ML-driven experience optimization.
- End-to-end ownership: discovery, IA, interaction design, research, prototyping, usability, delivery.
- Strong in design systems, accessibility, data visualization, and desktop-first enterprise apps.
- Master of Computer Application (M.C.A) at University of Madras, Chennai.
- Master of Science (M.Sc) in Psychology at University of Madras, Chennai.
- Pursuing Ph.D in Human-Computer Interaction/Immersive Technology at NITTTR, Chennai (2026-2029).

Career Objective

To consistently achieve a win-win for the organization and clients by enhancing product value and customer satisfaction. Seeking opportunities to update my knowledge and skills with the latest trends and be part of a team dedicated to organizational growth.

Professional Experience

HCL Technologies, Chennai, India June 2024 - Present	Designation: **Technical Lead, UX/Product Design**

Lam Research Corporation, United States https://www.lamresearch.com/

- Reduced digital asset search time by 35% and increased initial user adoption by 60% by directing user research and partnering with engineering to integrate an intelligent, LLM-driven Smart Search platform that maps semantic user intent across 14+ complex resource types.
- Improved search result relevance by 40% by applying cognitive psychology principles to information architecture, minimizing interaction friction through advanced filtering systems, and designing enterprise-grade, desktop-first UI components.
- Mitigated user choice paralysis and cognitive fatigue by architecting scalable, AI-powered predictive search capabilities and multi-variant metadata enrichment frameworks that perfectly adhere to strict corporate design system standards.

Cadent Gas Ltd, United Kingdom https://www.cadentgas.com/

- Achieved a 79% user satisfaction rate for high-stress operational safety monitoring by revamping a real-time gas emergency dashboard optimized via behavioral psychology frameworks to lower the mental workload of emergency response agents.
- Boosted gas leak detection rates by 20% while reducing operational emissions by designing intuitive user interfaces for the Digital Platform for Leakage Analytics (DPLA), successfully transforming predictive ML analytics into transparent, high-trust visualizations.
- Accelerated critical human-in-the-loop decision-making speeds by structuring optimized interaction hierarchies that translate complex, automated sensor data streams into actionable emergency reporting workflows.

Microsoft, United States https://www.microsoft.com/

- Elevated tech manager satisfaction by 30% during complex data center operations by serving as a key individual contributor to the redesign of the DCS-BSH Public Dashboard.
- Reduced task completion friction for server management and rebound technology workflows by synthesizing complex server tracking metrics into a cognitive-friendly, high-density live monitoring interface.

HCL America, United States May 2023 - March 2024	Designation: **Technical Lead, UX/Product Design**

PenFed Credit Union, United States https://www.penfed.org/

- Accelerated borrower completion rates by achieving a 25% reduction in mortgage application time by collaborating with cross-functional teams to map borrower pain points and streamline the end-to-end UX flow.
- Maintained 100% design-to-production consistency across high-volume sales funnels by authoring interactive component libraries and CMS design support natively tailored to the Salesforce design system.

HCL Technologies, Chennai. January 2021 - May 2023	Designation: **Technical Lead, UX/Product Design**

Synchrony Inc, United States https://www.synchronybank.com/

- Boosted cross-functional scheduling and design consistency by 30% by authoring and publishing consolidated UI usage guidelines that established uniform calendar design heuristics and aligned developer and product management mental models.
- Minimized user cognitive friction within a meeting rescheduling system by designing an intelligent, predictive interface that harmonized complex availability data streams, reduced visual clutter, and optimized interactive components to elevate digital engagement by 25%.

World Bank, United States https://www.worldbank.ore/

- Led teams and stakeholders through brainstorming sessions, requirements gathering, and business-logic definition for the International Development Association (IDA). Maintained a user-centered design perspective, resulting in a 20% increase in project efficiency.
- Designed efficient content models, security models, workflows, and templates, contributing to a 15% improvement in overall system performance.

Caterpillar Inc (Offshore, United States) https://www.caterpillar.com/

- Drove heavy machinery asset tracking efficiency and user satisfaction up by 20% by overhauling the interaction architecture for a real-time, live telemetry monitoring dashboard tracking dynamic vehicle positions and payload metrics during critical load and release cycles.
- Minimized operator cognitive load across diverse fleet deployments by applying behavioral psychology principles to a high-density dashboard, seamlessly organizing data streams for multiple truck types, fuel consumption, and predictive live monitoring analytics.

Prodapt Solutions Pvt.Ltd, Chennai. December 2018 - December 2020	Designation: **Lead Engineer, UX/Product Design**

CenturyLink (Offshore, United States) https://www.centurylink.com/

- Collaborated with clients and developer teams to design and improve UI, reducing complexity and enhancing

usability. Developed high-level and detailed storyboards, mockups, and prototypes to effectively communicate Interactions.

Emerio Technologies Private Ltd, Chennai. November 2017 - December 2018	Designation: **Sr. UI UX Developer**

- Created information architecture and designed outcomes-focused websites, applications, and widgets.
- Empathized with user experiences, defined user goals and problems, and established usability criteria.
- Developed storyboards, user personas, competitor analysis, user flows, conceptual mind mapping rapid prototypes/pencil sketches, and high-level prototypes.

Quadra Point Brand Services Pvt. Ltd, Chennai. April 2016 - October 2017	Designation: **Sr. Graphic & Web Designer**

- Designed web layouts and wireframes for applications, including descriptions and navigation. Created ads, brochures, flyers, package designs, new product labels, stickers, banners, and print media advertisements.

Imitor Graphica, Chennai. January 2015 - May 2015	Designation: **Presentation Specialist**

- Created impactful presentations (Prezi, PPT) for international clients, directly supporting company growth initiatives. Responsible for developing creative strategies and brand identities for a diverse portfolio of clients.

Sri Devar Pictures, Chennai. September 2011 - December 2014	Designation: **Graphic Designer & Desktop Engineer**

- Created new design themes for marketing and collateral materials. Designed corporate branding and identity for diverse products, and handled print and package design for multiple industries to various print formats.

Skills

- **Experience Design:** IA, journey mapping, task flows, interaction models, accessibility (WCAG), high/low fidelity prototypes.
- **Research Methods:** expertise encompasses user interviews, surveys, competitor analysis, heuristic evaluation, usability studies, participatory design, ethnography, metric analysis, PURE analysis, Wizard of OZ, literature review, thematic analysis, design thinking, problem solving, and creative and critical
- **Design & Prototyping Tools:** Figma, Sketch. Adobe XD, Photoshop, Illustrator, InVision, Marvel.
- **Research & Testing Tools:** SurveyMonkey and UserTesting.com
- **Project Management:** JIRA and Notion
- **Development Skills:** HTML5, CSS3, Bootstrap, and JavaScript (ReactJS, NodeJS - Beginner)

Certifications & achievements

- **Certifications:** Adobe Certified UX Designer, AEM Developer Certified, Government of India Certified Design Thinking Practitioner, IBM Creative/Critical Thinking & IBM Enterprise Design Thinking Co-Creator Badge, Google Creative Certification
- **Workshops & Training:** XR Workshop, IIT Madras | Focused training on Extended Reality (AR/VR/MR) design, spatial UX, and immersive prototyping.
- **Sports & Extracurriculars:** National & International Power 20 Cricket | HCL Chennai Cricket Team